

What is a Lyzr Agent

LESSON 01 · MODULE 1 — FOUNDATIONS

What you'll learn

- What a Lyzr Agent actually is in concrete terms
- How to install the Lyzr ADK and configure a single API key for the whole SDK
- The five parameters that define a basic agent
- How `agent.run()` returns text from a model
- Where every other capability (streaming, structured outputs, memory, tools, knowledge bases) plugs into this same shape

Key concepts

One API key, not one per provider. Lyzr handles provider credentials server-side. You sign in at `studio.lyzr.ai`, copy your API key from the account page, and put it in a `.env` next to your notebook. The SDK reads `LYZR_API_KEY` automatically — no OpenAI key, no Anthropic key, no per-vendor secret to wrangle.

Studio is the entry point. Two lines of setup: `from lyzr import Studio` and `studio = Studio()`. That's how you create agents, knowledge bases, and everything else the SDK exposes.

Five parameters define a basic agent. `name` (a label so you can find it later), `provider` (the LLM — e.g. `gpt-4o`), `role` (what the agent is), `goal` (what it's *trying to do*), and `instructions` (how it should respond). Memory, tools, knowledge bases, structured outputs — they all hang off the same `create_agent` signature, layered on top.

An agent is three things. An LLM, a set of instructions that shape how it responds, and a run loop (`agent.run()`) that ties them together. That's the whole mental model. Everything else in this course is a way to feed more context in, get more structured data out, or let the agent take action.

No ceremony. If you're coming from other agent frameworks, you might expect more scaffolding. There isn't any here, and that's the point.

Code shown in this lesson

BASH

```
pip install lyzr-adk==0.1.9
```

TAKEAWAY

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PYTHON

```
from lyzr import Studio
studio = Studio()
```

PYTHON

```
agent = studio.create_agent(
    name="my-assistant",
    provider="gpt-4o",
    role="A helpful assistant",
    goal="Answer the user's questions accurately",
    instructions="Be concise.",
)

response = agent.run("What is an agent?")
print(response.response)
```

TRY THIS

- 1 Re-run the agent in cell 4 with `provider="claude-sonnet"` instead of `"gpt-4o"`. Same question, same instructions. Compare the two answers side-by-side. What changed — the substance, the voice, both?
- 2 Change the instructions from `"Be concise."` to `"Answer in exactly three sentences."` Re-run. Verify the constraint held; if it didn't, note which sentence the model added or dropped.